TASK 8 – Trace the Compilation Process:

8.1 GCC stages and outputs:

```
# Preprocess: expand macros and include header contents
gcc -E main.c -o main.i

# Compile to assembly
gcc -S main.c -o main.s

# Compile to object file
gcc -c main.c -o main.o

# Link source files into an executable
gcc main.c tariff.c -o billing

# Enable useful warnings
gcc -Wall -Wextra -Wpedantic -std=c11 main.c tariff.c -o billing
```

Stage	Command / output	What happens
Preprocessing	gcc -E; .i file	Includes headers, expands macros and processes conditional compilation.
Compilation to assembly	gcc -S; .s file	Checks and translates C into assembly language.
Assembly	gcc -c; .o object file	Creates machine-code object files; external references may remain unresolved.
Linking	gcc main.c tariff.c -o billing	Combines object code and libraries into an executable.

8.2 Error categories:

- Compile-time error: a syntax error such as a missing semicolon or an undeclared identifier.
- Link-time error: calling a function declared but not defined in any linked source/object file, leading to an undefined reference or unresolved external.
- Run-time error: invalid input, arithmetic overflow in an unchecked calculation, or a program crash while executing. A logical error such as applying the wrong slab may run without crashing but still produce an incorrect bill.

Warnings from -Wall -Wextra -Wpedantic can reveal unused variables, suspicious conversions, format-specifier mismatches, or non-standard code. Resolve warnings rather than suppressing them without understanding the cause.

8.3 One file versus multiple files:

In a single-file program, changing any part normally requires recompiling the entire translation unit. In a multi-file project, only changed source files need to be recompiled, then the object files are linked again. The total object-file space is approximately the sum of the individual object files, plus build metadata. Splitting the program adds organisation and build steps but helps larger projects.

```
gcc -Wall -Wextra -std=c11 -c main.c -o main.o
gcc -Wall -Wextra -std=c11 -c tariff.c -o tariff.o
gcc main.o tariff.o -o billing
```